Control Strategies for Renewable Energy Sources and Battery Systems in Low-Voltage Networks

Special Subject
Maria Apostolidou

Supervisor :
Kontis Eleftherios
University of Thessaly
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Abstract

This study presents a comprehensive evaluation of control strategies for Renewable Energy Sources (RES) and Battery Energy Storage Systems (BESS), designed to mitigate voltage violations in Low-Voltage (LV) distribution networks characterized by high penetration of Photovoltaic (PV) systems. Using an integrated OpenDSS–MATLAB co-simulation environment, we analyze the performance of different inverter-based regulation techniques, including Q–V droop, P–V droop, and Power Factor (PF) control. The research extends to the integration of BESS units, employing a Constant Power (CP) charging/discharging framework to increase grid stability. Experimental results show that while traditional droop mechanisms effectively constrain voltage within the 0.95–1.05 p.u. limits, the combination of PF control with the existence of battery systems provides a more robust voltage profile. The study highlights the ability of decentralized controllers to prevent overvoltages or undervoltages in distribution network elements, offering a reliable voltage regulation solution.

1 Introduction

Nowadays, the need for clean and affordable energy has led to a radical restructuring of the way electricity networks operate [25]. Low-voltage (LV) networks , which traditionally transported energy from the central system to consumers, are now being transformed into active systems where users themselves generate and store electricity [15]. This change is necessary to address climate change, but it brings with it significant technical difficulties [14].

The main challenge lies in the instability of Renewable Energy Sources (RES), particularly Photovoltaic (PV) systems, due to their increasing penetration in LV distribution networks. High RES penetration may lead to voltage rise, phase unbalance, reverse power flows, and increased stress on distribution assets, especially in weak LV feeders, because their production depends on the weather and can change abruptly [6, 4, 19]. These issues need the deployment of advanced control strategies capable of ensuring grid stability while maximizing renewable energy usage. One of the most critical problems examined in this work is the existence of overvoltages and undervoltages in the grid. For example, high production from PV during hours of intense sunshine can "stress" the grid, causing overvoltages , while a sharp increase in demand or a drop in production can lead to undervoltages [19, 6].These phenomena, if left unchecked, can cause damage to consumer devices or even complete system instability [6].

This is where Battery Energy Storage Systems (BESS) can really help. Batteries act as regulators: they store energy when there is a surplus and deliver it to the grid when needs increase, helping to smooth out voltages [24, 9]. However, in order for RES and batteries to operate harmoniously and to maintain constant voltage on the buses, control strategies are required.

Inverter-based Distributed Energy Resources (DERs) play a important role in modern LV networks, as power electronic interfaces allow flexible control of active and reactive power injection. Among the available control solutions, local and decentralized strategies are particularly attractive due to their scalability, low communication requirements, and robustness against communication failures [6, 10]. Such approaches are widely adopted in microgrids and distribution networks, where centralized control may be impractical. Control strategies for inverter-based RES are commonly structured within a hierarchical control framework, which separates control tasks according to different time scales and functionalities.

- Primary control: Acts instantaneously to contain the voltage locally, using methods such as droop control [24, 6].

- Secondary control: Corrects deviations and restores voltage and frequency to their nominal values [24, 6].
- Tertiary control: Deals with overall financial management and optimal energy flow [10, 5].

The reaction time and operational complexity both rise when the control level moves from primary to tertiary [25]. Within this framework, RES control methods such as inverter-based control, maximum power point tracking (MPPT), active power curtailment, and droop control have been extensively studied as effective tools for voltage regulation and grid support in LV networks [6, 2, 19]. Due to their simplicity, droop and local control strategies are widely considered suitable for practical distribution network applications.

Droop control techniques have emerged as a fundamental solution for voltage and frequency regulation in energy systems. By imitating the behavior of synchronous generators, droop-based controllers enable proportional power sharing and autonomous operation without relying on communication infrastructure [21, 25, 17]. In LV networks, droop control is commonly applied through P–f, P–V and Q–V, allowing PV inverters to support voltage regulation under varying operating conditions.

BESS are useful for supporting RES integration and enhancing operational flexibility in LV networks. BESS control is commonly organized around battery management system (BMS) functions, state-of-charge (SoC) regulation, and charging and discharging control strategies [1, 9, 3]. Conventional charging methods such as Constant Current (CC), Constant Voltage (CV), and CC–CV are widely adopted due to their robustness and proven impact on battery lifetime [1, 3, 22]. In parallel, Constant Power (CP) charging and discharging strategies have gained attention in grid applications, as they allow predictable power exchange with renewable generation [8, 20]. The selection of an appropriate control strategy therefore plays a critical role not only in battery health but also in the overall dynamic behavior of distribution networks.

This work investigates control strategies for RES and BESS in LV networks through a comprehensive study. A realistic LV network is modeled in OpenDSS and co-simulated with MATLAB. Initially, for the PV systems several control strategies are evaluated, namely Q–V droop control, P–V droop control, power factor (PF) control and a no control approach to understand the deviations. Subsequently, for the PF-controlled PV scenario, CP charging and discharging strategy is examine. The comparative analysis provides insights into the impact of coordinated RES and BESS control on voltage regulation and system performance in LV distribution networks. The final aim is to find out how the voltage is affected, using different voltage regulation methods for the PV systems and the effect of battery systems existence working under a charging/discharging method. The remainder of this report is structured as follows: Section 2 reviews the relevant literature on PV and BESS control, Section 3 describes the distribution network under study, Section 4 outlines the proposed methodology and control strategies, Section 5 details the experimental framework, while Sections 6 and 7 present the simulation results and the final conclusions of the study.

2 Related Work

This section reviews prior research relevant to the control strategies adopted in this work, focusing on inverter-based PV control methods and battery charging/discharging strategies in LV networks and microgrids. Emphasis is placed on studies that employ concepts similar to those investigated in this paper.

2.1 PV Inverter Control in LV Networks

High penetration of PV systems in LV networks has motivated extensive research on inverter-based voltage regulation techniques. The electric power generated from several renewable resources is in direct current (DC) form and converted to alternating current (AC) by an inverter [21]. Inverter-based voltage regulation is the process where smart power electronics stabilize the grid by dynamically adjusting the injection or absorption of reactive power and real power based on local voltage measurements. Reviews such as [6, 4, 19] highlight that local inverter control strategies are among the most practical solutions for decreasing voltage rise and power quality issues.

Several studies investigate the impact of reactive power control using Q–V droop characteristics. In [21, 25, 17], droop-based reactive power control is shown to effectively regulate voltage and share reactive power among distributed inverters in microgrids. Similar approaches are discussed in [10, 15], where droop control forms the primary control layer within hierarchical microgrid architectures. These works demonstrate that Q–V droop control is particularly suitable for LV networks, where voltage sensitivity to reactive power is high.

Active power voltage regulation has also been explored through P–V droop control and active power curtailment strategies. The study in [2] analyzes active power curtailment as a planning and operational tool to increase RES capacity, showing that controlled curtailment can significantly reduce overvoltage events. Moreover, [25] reports that hybrid droop schemes combining active and reactive power control can further enhance voltage stability under high RES penetration.

PF method remains a common practice in distribution networks due to grid requirements and implementation simplicity. According to [6, 19], PF control is often used as a baseline scenario in comparative studies, although its limited adaptability restricts its effectiveness under dynamic operating conditions. These observations motivate its inclusion as a reference case in the present work.

2.2 Battery Systems for Grid Support

Battery Systems have been widely studied as flexible assets for supporting RES integration and enhancing grid stability. Foundational works such as [9, 7] provide detailed modeling approaches for BESS in the grid, while applications in microgrids and LV networks are extensively discussed in [11, 18]. These studies demonstrate that BESS can effectively decrease RES unexpected production, smooth power fluctuations, and contribute to voltage and frequency regulation.

Several works investigate coordinated RES and BESS operation. In [11], intelligent control of BESS is applied to smooth renewable power output, while [5] uses model predictive control for optimal power flow in unbalanced microgrids. Although these approaches achieve good performance, they often rely on centralized optimization and communication, which may limit effectiveness in practical LV networks.

2.3 Battery Charging and Discharging Control Strategies

Battery charging control has a direct impact on efficiency, thermal behavior, and lifetime. Classical strategies such as Constant Current (CC), Constant Voltage (CV), and CC–CV are extensively reviewed in [1, 3], where their advantages and limitations are discussed in the context of renewable energy systems. These methods are widely adopted due to their robustness and ease of implementation.

Advanced charging strategies have also been proposed to optimize charging time and battery degradation. In contrast to traditional CC–CV methods, multi-stage and optimized current profiles are examined in [12, 13, 23], showing enhanced thermal performance and longer battery lifetime. Although effective, such approaches increase control complexity and are often designed for autonomous charging systems rather than BESS integrated with the grid.

Constant power charging and discharging strategies have been explored primarily in DC microgrid and power electronics applications. In [8] and [20], constant power operation is analyzed and compared with constant current control, showing that both methods can achieve similar efficiency under specific boundary conditions. These findings suggest that constant power control can be attractive for grid applications where predictable power exchange is required.

2.4 Control Strategies used in this work

2.4.1 PV Control Methods

- **No control strategy**

  The no-control operating mode represents the baseline case in which photovoltaic inverters inject active power at the maximum available level, typically determined by the maximum power point tracking (MPPT) algorithm, without providing voltage or reactive power support. While this approach maximizes local energy production, it may lead to voltage rise and power quality issues in LV feeders with high RES penetration [6, 4, 19]. Consequently, no-control operation is mainly used as a reference scenario for assessing the effectiveness of advanced inverter control strategies.

- **Q–V Droop Control**

  Reactive power–voltage (Q–V) droop control is one of the most widely adopted grid-supporting strategies for inverter-based DERs. In this method, the inverter adjusts its reactive power output proportionally to local voltage deviations [21, 25, 17]. Q–V droop control is particularly effective in LV networks, where voltage sensitivity to reactive power is significant. Numerous studies have demonstrated that reactive power control through droop characteristics can mitigate overvoltage conditions and improve voltage profiles under high PV penetration [6, 19].

- **P–V Droop Control**

  Active power–voltage (P–V) droop control regulates the injected active power as a function of local voltage magnitude. When voltage exceeds predefined thresholds, the inverter curtails active power output to prevent further voltage rise [2, 25]. Although this method directly impacts power, it provides a robust mechanism for voltage regulation in scenarios where reactive power capability is limited. Active power curtailment strategies are increasingly considered in distribution network planning as a means to increase RES hosting capacity while preserving network constraints [2].

- **Power Factor Control**

  Power factor control or $\cos\phi(P)$ is a local voltage regulation method, in which the power factor of the converter is dynamically adjusted as a function of the instantaneous injected active power. Unlike the constant power factor method, this approach is activated when the active power generation exceeds a certain threshold, which is typically set at 50% of the unit's rated power.When the output exceeds this limit, the inverter gradually reduces the power factor (usually down to 0.90 or 0.95 inductive), going into under-excited mode.Through this process, the photovoltaic system absorbs reactive power from the grid, thus contributing to the voltage rise mitigation caused by the high injection of active power into the distribution networks [16].

2.4.2 BESS Charging and Discharging Strategy

- **Constant Power Control**

  CP charging and discharging regulates the battery power exchange with the grid at a fixed level, allowing both current and voltage to vary dynamically. CP control is particularly attractive for grid-support applications, as it enables predictable power flow management and helps coordination with RES generation [8, 20]. Studies comparing CP and CC strategies have shown that both methods can achieve similar efficiency under certain operating conditions, although their impact on battery stress and grid interaction differs [20].

The reviewed literature demonstrates that PV inverter droop control and BESS charging/ discharging strategies have been extensively studied as separate research topics. Several works evaluate droop-base voltage regulation for RES, while others focus on battery charging optimization and lifetime enhancement. However, fewer studies provide a unified assessment of how different PV inverter control strategies interact with BESS charging and discharging methods within realistic LV networks. In contrast to many existing works that emphasize either advanced optimization or standalone battery performance, this study tries to focus on a comparative evaluation of commonly deployed control strategies. By combining PV control and a BESS charging/discharging method within an OpenDSS–MATLAB co-simulation framework, this work aims to provide practical insights into the impact of RES and BESS control in LV networks.

3 Distribution Network Under Study

For all the analyses and scenarios examined in this project, the distribution network shown in Fig. 1 was used. It is a LV network that follows the operating standards of the European Union and is located in Romania.

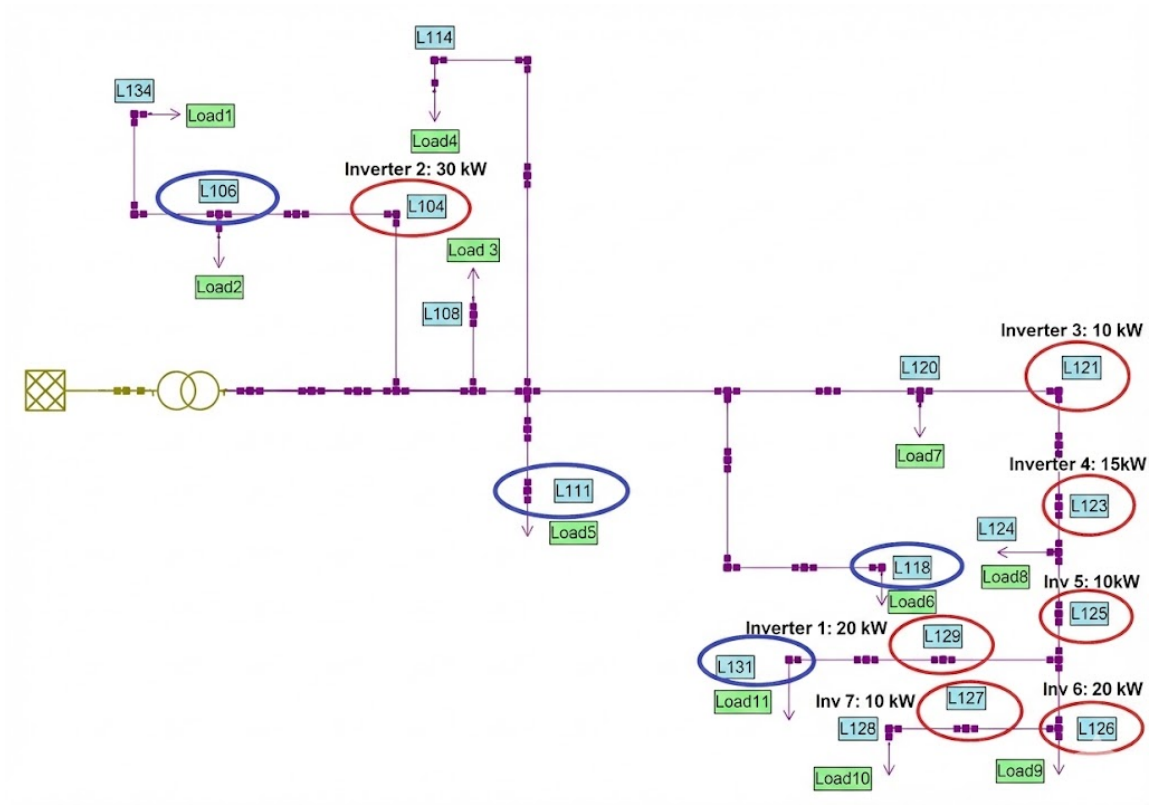


Figure 1: Distribution network under study.

The distribution network operates at a base frequency of 50 Hz and is structurally defined by 36 buses interconnected via 35 lines with uniform resistance and reactance (0.215/0.334 Ω/km). Voltage is stepped down from the 20 kV medium-voltage source to the 0.69 kV low-voltage buses through a 630 kVA Delta-Wye transformer bank with 4.6% impedance. The network includes 11 loads, whose total consumption and connection buses are reported in Table 1. As shown, the loads are unbalanced three-phase and the values are given in kW and kVAr for active and reactive power. In addition, 7 three-phase PV units are connected through inverters, with their rated active power and locations listed in Table 2. Finally, 4 three-phase BESS units are installed, each unit is characterized by its rated active power and energy capacity and operates at a nominal DC pack voltage of 400 V, and is constrained within a SoC range of 15%–95% with an initial SoC of 50%, while their ratings and connection buses are summarized in Table 3.

Table 1: Loads per phase and locations.

Loads	P_A (kW)	Q_A (kVAr)	P_B (kW)	Q_B (kVAr)	P_C (kW)	Q_C (kVAr)	**Bus**
Load1	4.20	1.26	0.00	0.00	2.52	0.84	L134
Load2	5.88	1.68	2.52	0.84	5.04	1.68	L106
Load3	5.04	1.68	2.52	0.84	5.88	1.68	L108
Load4	5.04	1.68	4.20	1.26	5.04	1.68	L114
Load5	3.36	1.68	2.52	0.84	4.20	1.68	L111
Load6	2.52	0.84	0.84	0.42	3.36	1.68	L118
Load7	2.52	0.84	0.00	0.00	2.52	0.84	L120
Load8	2.52	0.84	0.00	0.00	0.00	0.00	L124
Load9	2.52	1.68	3.36	1.26	2.52	0.84	L126
Load10	2.52	0.84	0.00	0.00	0.00	0.00	L128
Load11	3.36	1.26	2.52	0.84	1.68	0.84	L131

Table 2: PV units power ratings and locations.

PV	P (kW)	**Bus**
PV1	20	L129
PV2	30	L104
PV3	10	L121
PV4	15	L123
PV5	10	L125
PV6	20	L126
PV7	10	L127

Table 3: BESS key parameters and locations.

BESS	P_{rated} (kW)	E_{cap} (kWh)	**Bus**
BESS1	20	80	L106
BESS2	15	60	L111
BESS3	20	80	L118
BESS4	15	60	L131

Load and PV generation profiles The power output of the PV units, as well as the power demand of the network loads, varies over time and does not remain fixed. Fig. 2 illustrates the common PV production profile used for all PV systems in the considered topologies. In a same way, Fig. 3 depicts the load demand profile applied to all loads. In essence, these profiles express the percentage of the rated power of each PV unit or load that is generated or consumed, at every time step.

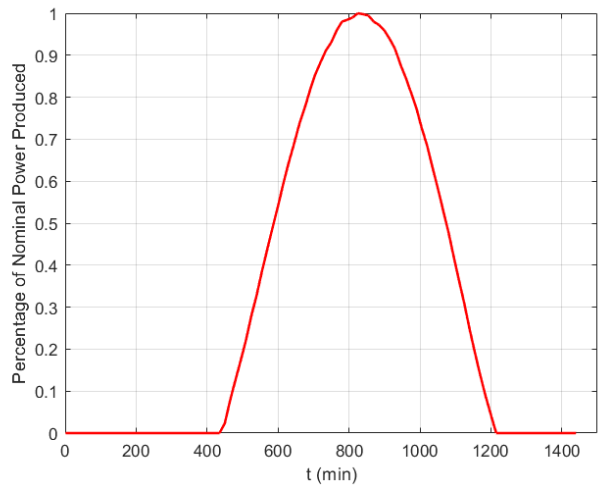


Figure 2: Generation profile.

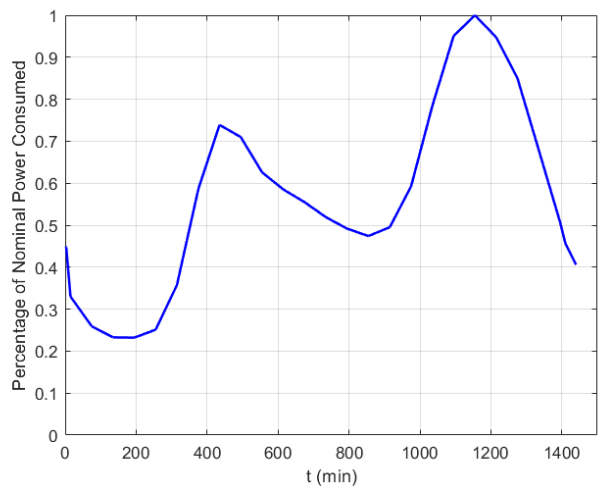


Figure 3: Consumption profile.

4 Methodology

All simulations were carried out in MATLAB version R2023b, with the distribution network modeled and solved using Open Distribution System Simulator (OpenDSS), an open source software used for simulations and analyses of electrical power distribution systems, in a time-series framework. The experimental strategy was organized in two stages.

4.1 Stage I: PV Control

This stage examines the impact of each strategy on voltage regulation and overall network performance under the same loads and PV generation conditions. These are the four operating cases that were evaluated:

- **No Control**: PV inverters operate without any control.
- Q–V **Control**: As shown in Fig. 4, the system operates within a central deadband where no action is taken ($Q = 0$), but if the voltage deviates significantly from the nominal 1 p.u. value, the inverter intervenes. Specifically, it injects reactive power to support the grid during undervoltage conditions and absorbs reactive power to decrease overvoltages, with the intensity of the response increasing linearly until the predefined limits V_{min} and V_{max} are reached. The function is described as:

$$Q(V) = \begin{cases} 1, & V < V_{\min}, \\ \dfrac{V - \left(1 - \dfrac{dV}{2}\right)}{V_{\min} - \left(1 - \dfrac{dV}{2}\right)}, & V_{\min} \leq V < 1 - \dfrac{dV}{2}, \\ 0, & 1 - \dfrac{dV}{2} \leq V < 1 + \dfrac{dV}{2}, \\ -\dfrac{V - \left(1 + \dfrac{dV}{2}\right)}{V_{\max} - \left(1 + \dfrac{dV}{2}\right)}, & 1 + \dfrac{dV}{2} \leq V < V_{\max}, \\ -1, & V \geq V_{\max}. \end{cases}$$

$$\begin{aligned} Q(V) &: \text{ target reactive power [kVar]}, \\ V &: \text{ actual terminal voltage in p.u.}, \\ V_{\min} &: \text{ lower voltage limit in p.u.}, \\ V_{\max} &: \text{ upper voltage limit in p.u.}, \\ dV &: \text{ deadband width in p.u. }, \end{aligned}$$

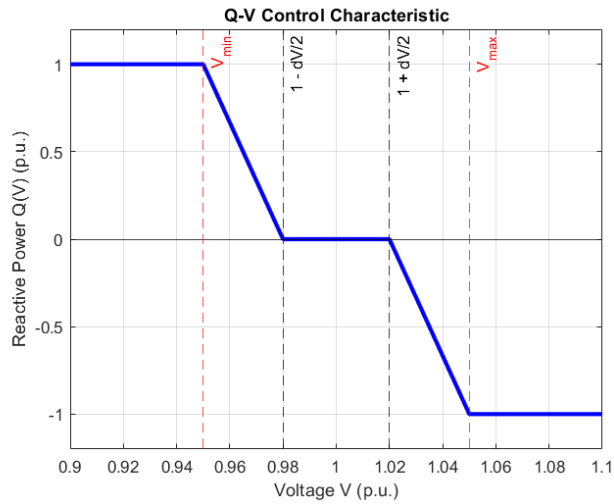


Figure 4: Q(V) Control.

- P–V **Control**: This method adjusts active power output as a function of the local voltage magnitude to ensure grid stability. The inverter maintains maximum nominal power as long as the voltageis below the threshold V_1, but once this limit is exceeded, it begins to linearly decrease power production. If the voltage continues to rise and reaches V_2, the active power is cut to zero, as shown in Fig. 5. The function is described as:

$$P(V) = \begin{cases} P_n, & V < V_1, \\ P_n \dfrac{V - V_2}{V_1 - V_2}, & V_1 \leq V \leq V_2, \\ 0, & V > V_2. \end{cases}$$

$P(V)$: target active power [kW],
V : actual terminal voltage in p.u.,
V_1 : lower voltage threshold in p.u.,
V_2 : upper voltage threshold in p.u.,
P_n : nominal maximum active power at the current step [kW],

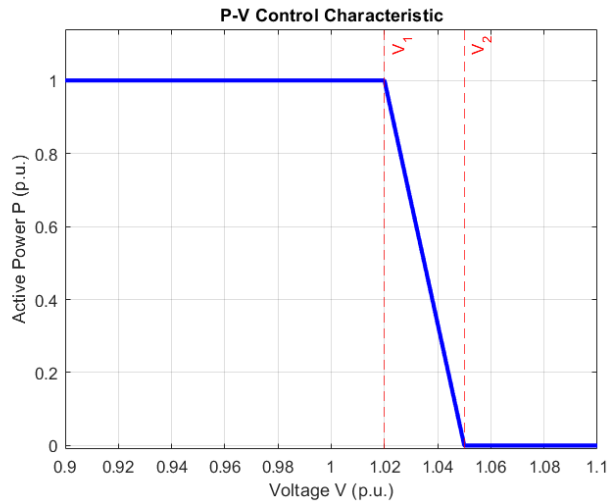


Figure 5: P(V) Control.

- **PF control**: It adjusts the inverter's power factor based on its active power output to support grid regulation. Following the control function bellow, the power factor remains at unity while production is below the 0.5 p.u. threshold, but as power production increases beyond that point, the power factor decreases linearly until 0.95, as shown in Fig. 6.

$$pf(P) = -0.1\,P + 1.05$$

pf : power factor.

P : percentage of nominal active power produced at a given time.

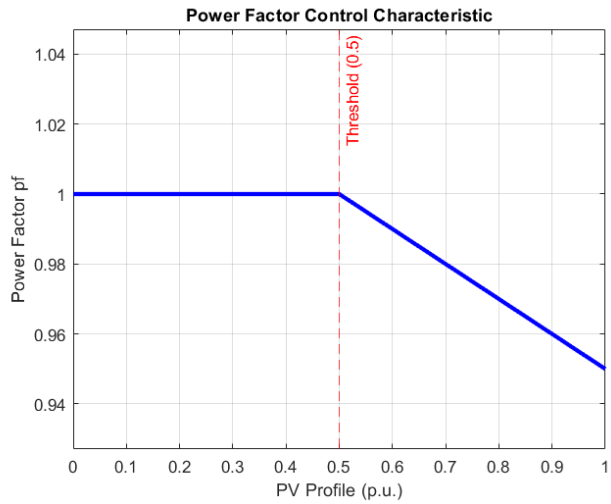


Figure 6: PF(P) Control.

4.2 Stage II: Battery operation under PF control

In the second stage, the PF case was extended by activating the BESS, which previously were assumed idle. The BESS units were allowed to charge and discharge in order to support the network operation. CP charging/discharging control mode was examined.

The objective of this stage is to observe how the existence of batteries under control, influence the behavior of the system, when combined with PF-based PV inverter operation.

5 Experimental Framework

The simulation follows a time-series approach to evaluate the impact of different control strategies on a LV network. The process is implemented using a co-simulation environment where MATLAB manages the control logic and OpenDSS runs the power flow.The methodology is structured around a central execution loop that processes data at one-minute intervals for a 24-hour period (1,440 time steps). At each time step, the system updates the load demand and solar PV generation profiles based on pre-loaded interpolated data. These values are injected into the OpenDSS circuit model. The script evaluates the current state of the grid to determine the necessary response from DERs. Depending on the selected configuration, the controller implements one of several strategies: (i) No Control, (ii) Q–VControl, (iii) P–V Control, (iv) PF Control.

For the second stage scenario, a BESS is working under a charging/ discharging logic. Battery operation is enabled only in the PF case. A simple schedule is adopted, where batteries charge between 10:00–16:00 and discharge between 18:00–22:00, otherwise remaining idle, while strictly maintaining the SoC within safe operational limits (15%–95%). Once the control commands are set, OpenDSS solves the non-linear power flow equations. The resulting phase voltages and positive sequence values are extracted back into MATLAB for stability analysis and visualization. To do a efficient analysis of the network performance, the post-processing stage includes a detection algorithm of the most critical bus. This selection process is based on two modes:

- **Positive Sequence Mode:** The code scans the positive sequence voltage (V_1) across all

buses, identifying the node with the highest cumulative duration of limit violations. The positive-sequence voltage component V_1, where $a = e^{j\frac{2\pi}{3}}$, is calculated as:

$$\vec{V}_1 = \frac{1}{3}(\vec{V}_a + a\vec{V}_b + a^2\vec{V}_c) \tag{1}$$

- **Phase Voltages Mode:** The check is extended to individual phase voltages (V_a, V_b, V_c). The bus exhibiting the longest violation duration on any single phase is selected.

Voltage stability was evaluated against the standard LV operational band, with violations defined as follows:

$$V > 1.05 \text{ p.u.} \Rightarrow \text{Overvoltage (OV)}, \qquad V < 0.95 \text{ p.u.} \Rightarrow \text{Undervoltage (UV)}. \tag{2}$$

The worst bus was selected according to the following rule:

1. If at least one OV event exists in the network, the worst bus is the bus with the longest OV duration.
2. Else, if at least one UV event exists, the worst bus is the bus with the longest UV duration.
3. Else, in the absence of any violations, the worst bus is identified as the one with the lowest stability margin.

6 Experimental Results

In this section the results, that were exported from the MATLAB and OpenDSS simulations will be presented. For each time step of the daily simulation, the phase-to-neutral voltages (V_a, V_b, V_c) were extracted at every bus and converted to per-unit using the base voltage $V_{\text{base}} = 690/\sqrt{3}$ V. In addition, the positive-sequence component V_1 was computed from the phase voltages. The slack bus was excluded from the scan. For the simulation with `nominal_mult = 1`, the network does not exhibit any OV/UV as shown in Fig. 7. Nominal multiplier is a scaling factor that uniformly increases or decreases all load and PV power levels to simulate different operating conditions. This is the occasion where the PV operate with the power ratings shown in Table 2. To see more clear results the `nominal_mult` was increased to 2.5, so the OV is obvious. The simulation evaluates the four PV control scenarios and the Combined PF Control with BESS logic. Finally, the BESS performance is validated by monitoring the SoC characteristic across the four storage units to ensure the charging/discharging logic is working effectively.

6.1 PV Control Strategies

Firstly this subsection compares the four PV operating modes ignoring the batteries. In all cases, the plotted signals correspond to the worst bus: the three phase voltages (V_a, V_b, V_c), the positive-sequence magnitude (V_1), and the limits at 0.95 and 1.05 p.u.

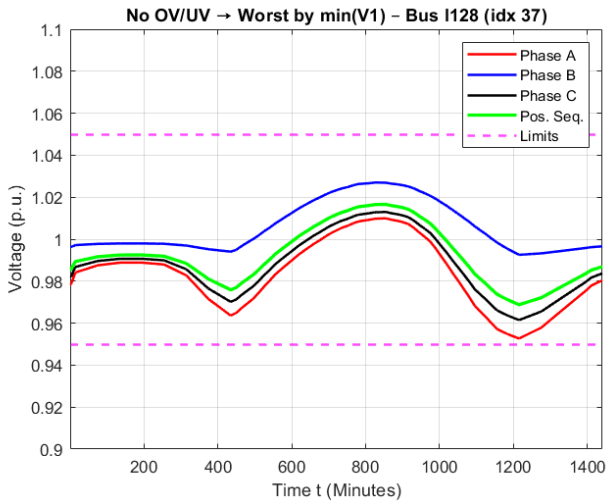


Figure 7: No control with the original PV power ratings.

6.1.1 No Control

Fig. 8(a) shows the basic operation where PV units inject active power according to the PV profile while operating at unity power factor ($Q = 0$). The daily voltage pattern follows the expected behavior of a LV feeder with distributed PV: voltages rise during high PV production (midday) due to local power export and decrease during periods of higher net load (morning/evening).

6.1.2 Q–V Control

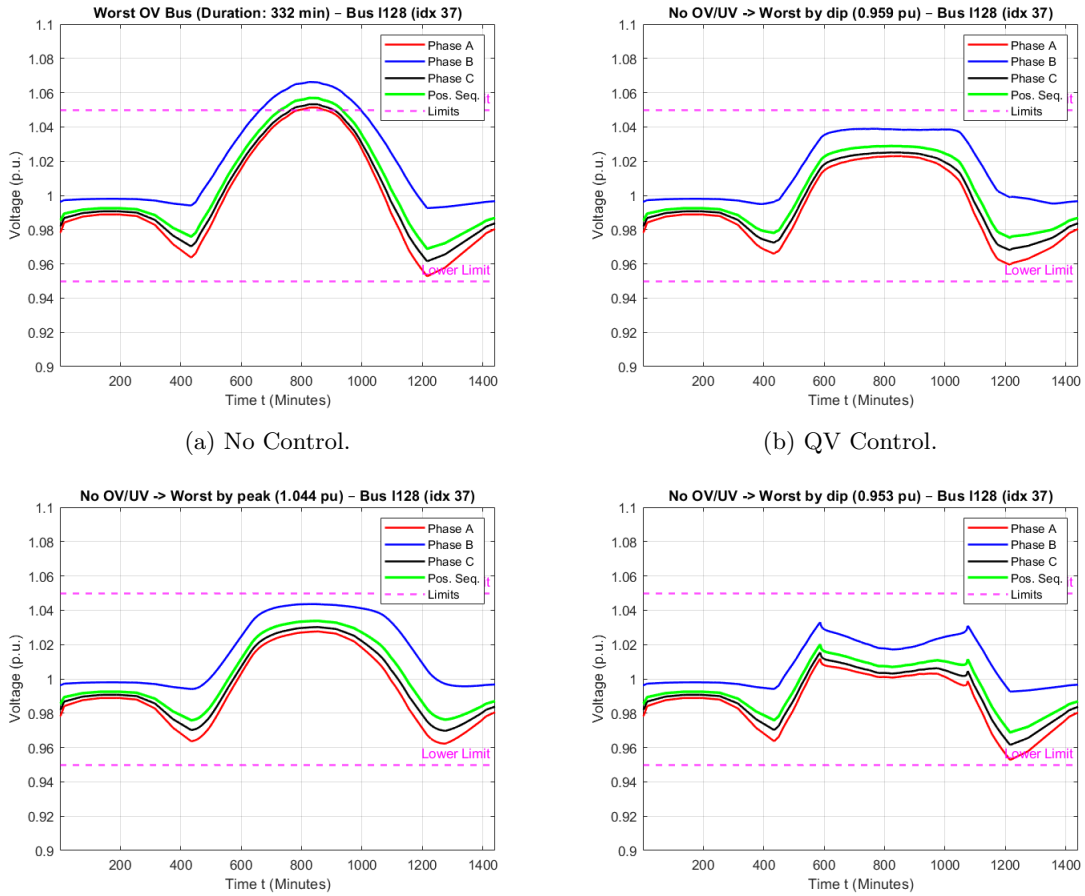
In Fig. 8(b), the $Q(V)$ Droop control modulates reactive power based on local voltage measurements to provide voltage regulation. The control regulates the voltage by absorbing reactive power when it exceeds the upper limit. When the active power causes the voltage to rise and exceed the limits, the inverter operates in an inductive mode by absorbing reactive power, which effectively lowers the voltage at the point of common coupling without requiring any reduction in active power production.

6.1.3 P–V Control

P–V Control (Fig. 8(c)) gradually reduces active power output when the grid voltage exceeds the 1.02 threshold. If the voltage continues to rise and exceeds the second critical threshold equal to 1.05, the controller completely cuts off the power output to prevent grid overload.

6.1.4 PF Control

Fig. 8(d) shows that when energy production increases, the controller reduces the power factor to force the system to absorb more reactive power. This automatic adjustment decreases the voltage rise caused by high production, thus keeping it within allowable limits. The sharp changes in the curves indicate the moments when the controller changes the power factor to adapt to the output.



(a) No Control.

(b) QV Control.

(c) PV Control.

(d) $\cos\varphi(P)$ Control.

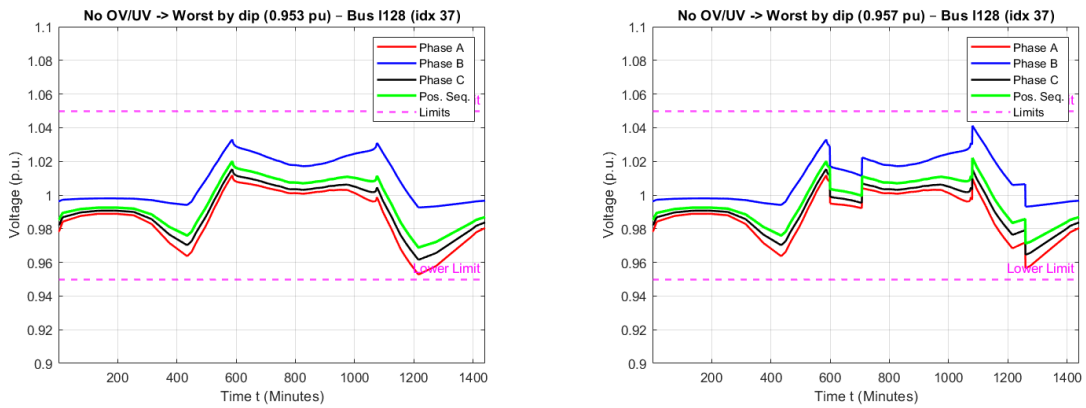
Figure 8: Worst-bus voltage characteristics for PV control strategies based on positive sequence voltage.

6.2 Battery-assisted operation under power factor control

After identifying $\cos\varphi(P)$ as a effective control strategy for voltage management, batteries are introduced to further shape the network power balance seen by the feeder. In all battery cases, the PV inverters continue to operate under $\cos\varphi(P)$ control, while the four storage units are working under CP method.

6.2.1 CP charging/discharging method

Fig. 9(b) presents the CP case versus Fig. 9(a) that shows $\cos\varphi(P)$ control. In the graph for this method, we notice that the curve is different due to energy storage. The sharp changes that appear are due to the moments when the batteries start or stop charging/discharging. Although, as mentioned above about the hours of charging and discharging, it is clear that the charging stops around minute 707 instead of 960. The discharging mode also stops at 1270 instead of 1320,



(a) $\cos\varphi(P)$ control.

(b) CP method.

Figure 9: Worst-bus voltage profiles under $\cos\varphi(P)$ PV control vs under $\cos\varphi(P)$ PV control and CP charging/discharging method for batteries.

because the SoC limits were reached. The performance of the CP charging strategy is illustrated by the SoC evolution over time in Fig. 10. During the charging, the SoC exhibits a strictly linear increase, which is characteristic of a constant energy injection rate consistent with fixed active power control (P = constant). Notably, the response curves for all four battery units overlap perfectly, demonstrating that the CP method maintains identical charging rates and perfect synchronization across the parallel storage units, due to their same initial SoC. The process effectively saturates once the SoC reaches the upper safety limit of 0.95, preventing overcharging before the subsequent discharge cycle.

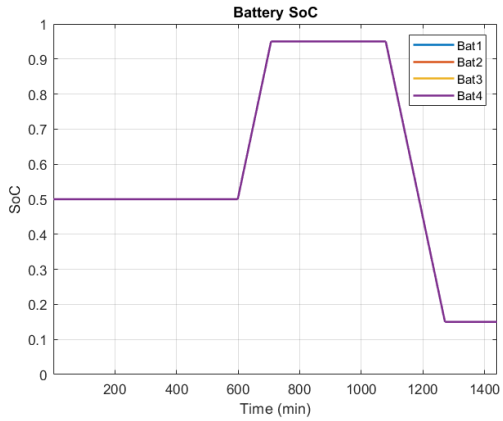


Figure 10: Battery SoC during CP Method.

7 Conclusion

In order to overcome the technical difficulties of reverse power flows and voltage imbalance, the incorporation of RES into traditional LV networks requires control frameworks. This study showed, that under no control may be extended overvoltage events during times of peak irradiance in this network, surpassing the allowable 1.05 p.u. threshold for roughly 332 minutes in the scenarios under investigation.

Comparative research shows that while P–V droop control offers a clear safety buffer against overvoltages, the required active power curtailment leads to non-ideal usage of RES. On the other hand, Q–V droop and PF strategies operate in a more profitable way. The integration of BESS units running under CP strategy resulted in the most suitable performance improvement. By absorbing excess active power, the BESS can successfully reduced voltage peaks, changing the net load profile. Nevertheless, the simulation also demonstrated that the effectiveness of BESS is also dependent on the availability of SoC. The duration of grid support is limited when the 95% SoC upper bound is reached during charging phases or the 15% lower bound is reached during discharging. In conclusion, a very effective way to update distribution networks and guarantee operational reliability under high RES penetration is to use a multi-layered control approach that combines BESS active power management and inverter reactive power modulation.

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